

Experiment # 04

Introduction to Proteus software with examples.

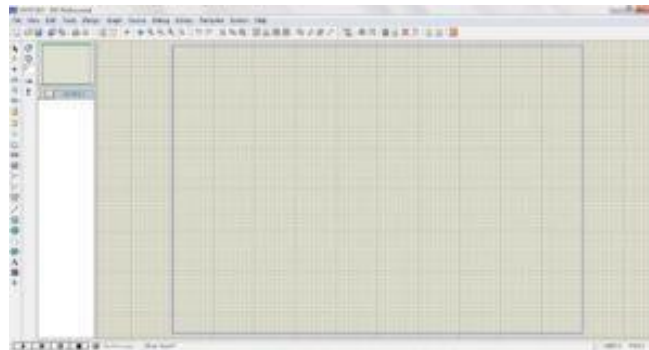
Objective

1. Familiar with Proteus software
2. Design circuit and their PCB Layout on Proteus

Software

- Proteus

Theory



- **Proteus Design Suite** (designed by **Labcenter Electronics Ltd.**) is a software tool set, mainly used for creating schematics, simulating Electronics & Embedded Circuits and designing PCB Layouts.
- Proteus ISIS is used by Engineering students & professionals to create schematics & simulations of different electronic circuits.
- Proteus ARES is used for designing PCB Layouts of electronic circuits.
- It's available in four languages i.e. English, Chinese, Spanish & French.

Why use Proteus ?

"Our circuit is working perfectly on Proteus but when we have implemented it on hardware, it's

not working." I receive a lot of such questions from engineering students, that's why, I am explaining what's the real purpose of Proteus:

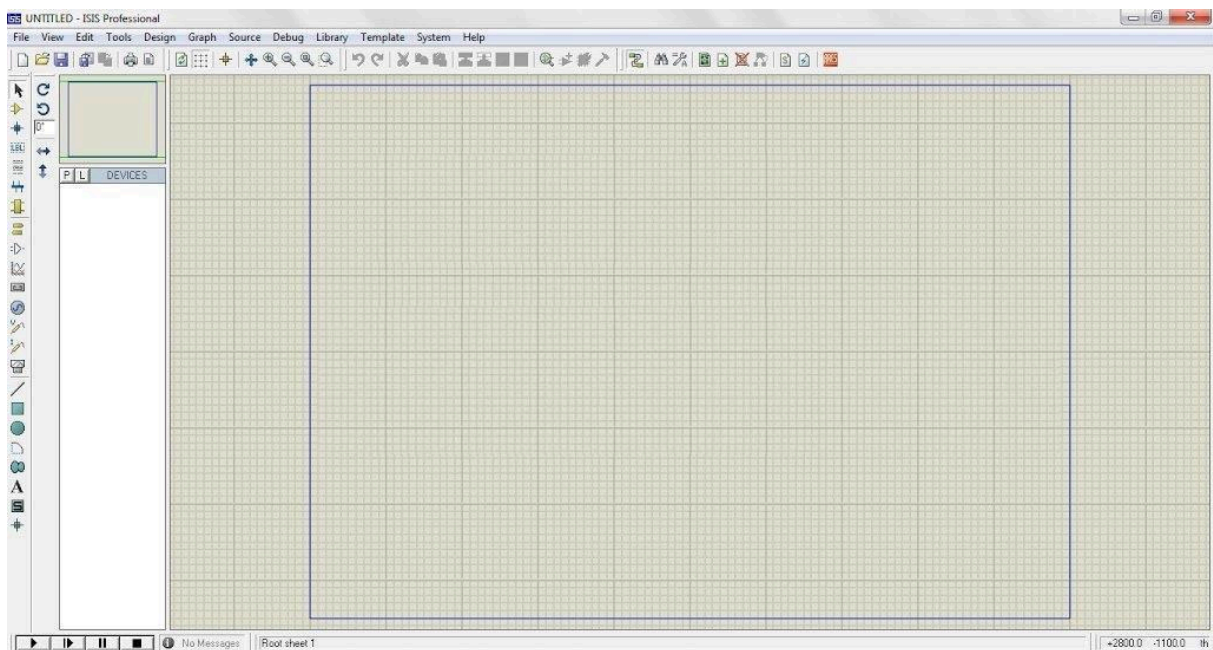
- Proteus is quite lenient in circuit designing and it works on ideal conditions i.e. if you don't add pull up resistors in Proteus simulation, then it won't give garbage value.
- Proteus is also used for PCB designing, we use **Proteus ARES** for that.
- Proteus is also used for designing/testing programming codes for different Microcontrollers i.e. Arduino, PIC Microcontroller, 8051 etc.
- In Embedded projects, we need to design a programming code for Microcontrollers and for designing such codes you have to perform a lot of testing, which involves uploading code to Microcontroller. So, in such projects, Proteus is a great relief. Let's say, you have

to print some strings on 20x4 [LCD](#), then it's quite annoying to burn the Microcontroller several times for typographical errors. Instead, design a circuit in Proteus and test your code in the simulation and once you are sure that you are getting perfect output then burn your PIC Microcontroller and test it on real hardware. Quite easy and handy. In the coming classes, I will show you how to burn code in Microcontrollers in Proteus.

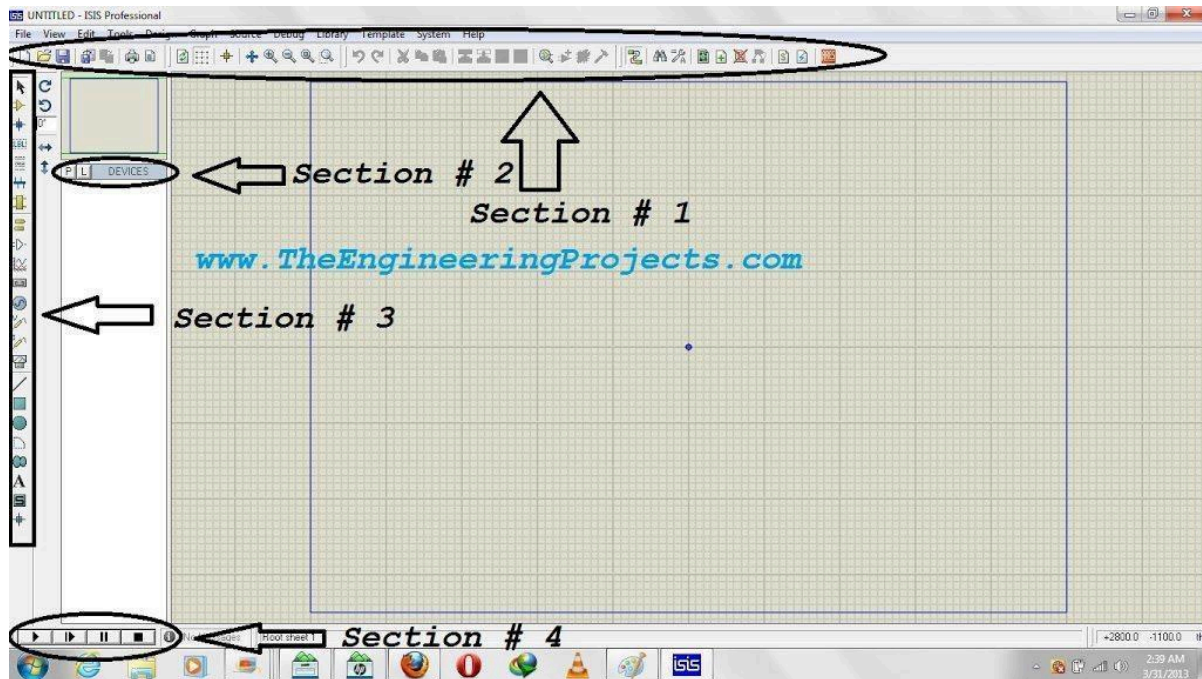
Getting Started with Proteus

You can download Proteus software from its [official website](#). So now I hope you have installed Proteus and ready to work on it:

- Click on Proteus ISIS and it will open as shown in below image.
- In the central area surrounded by blue lines, we design our circuit i.e. place the components and then join them together.



- As you can see in above figure that we have a lot of icons in Proteus software, so let's first understand these sections one by one.
- In the below image, I have divided the Proteus front-end in four sections:

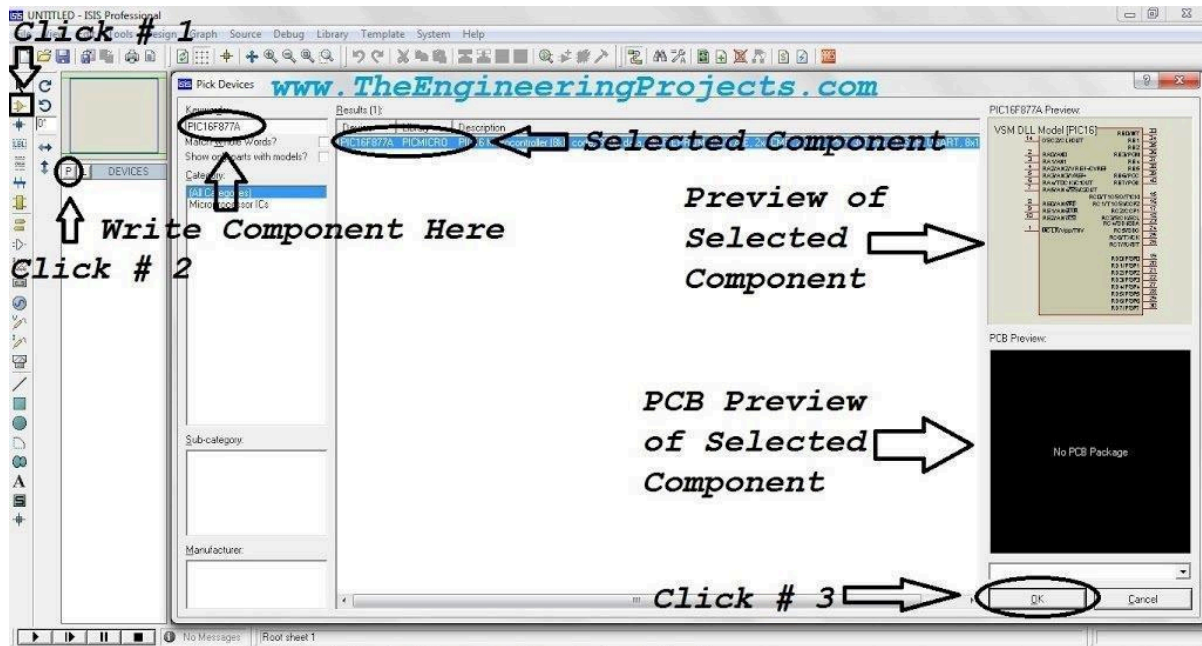


- **Section 1** is a toolbar which you would have seen on many simulation software, it has simple functionalities i.e. first icon to create a new layout, second one to open an existing layout, next one is to save layout, then there comes few zooming options and few other tools which we will discuss in coming tutorials.
- **Section 2** has two buttons. P is used to open the components list and E is used for editing purposes, like you want to edit the properties of any component then simply click on that component and then click on E and it will open the properties of that component and you can easily edit it.
- **Section 3** has different tools, used for designing circuits, we will discuss them in detail, at the end of today's tutorial.
- **Section 4** is the remote control section of Proteus, as it contains four buttons i.e. Play, Step, Pause & Stop. In order to run the simulation, we have to click on this play button.

Component Selection in Proteus ISIS

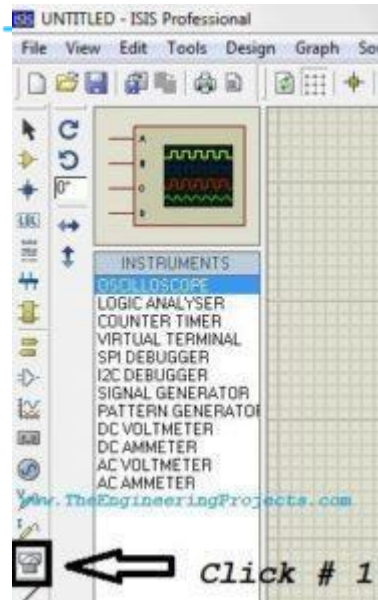
- As shown in below image, click on the icon that says Click # 1, it's a Component Mode Icon.

- After that click on P button and a new window will open up named Pick Devices.

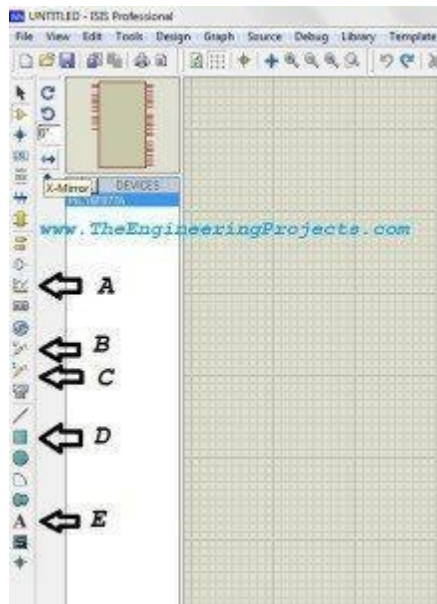


- In this new window there's a textbox on which Keyword is written, this text box is used for the component search.
- Proteus database has unlimited components in it so now in order to get your desired component, you have to search for it as I did.
- I have searched for AT89C51 and Proteus provided me that component along with its preview in top right corner and PCB package (if available).
- In order to add the component in Proteus workspace, either double click on it or click on the OK button.

Instruments in Proteus ISIS

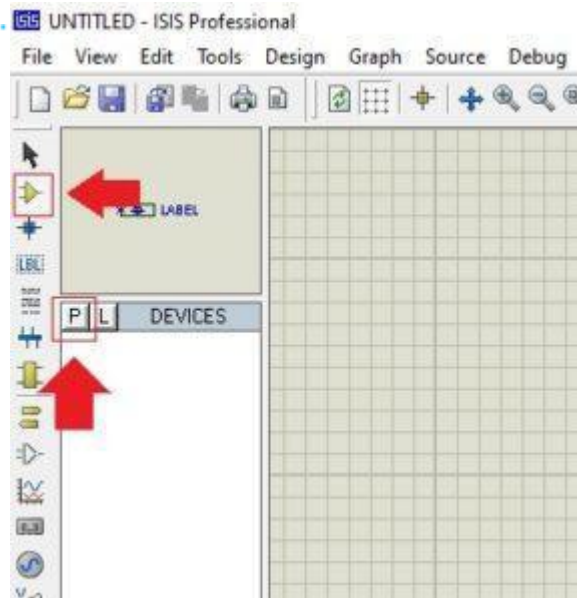


- There are few measuring instruments available in Proteus, which you can open by clicking the Instruments Icon, as shown in figure on right side.
- First one is [oscilloscope](#), we use it for viewing the behavior of different signals generated.
- Another important instrument is [Virtual Terminal](#), it is shown on the fourth number. This Virtual Terminal is used for checking data coming through [Serial Port](#).
- Then there's Signal Generator, it is used to generate signal like [sine wave](#) of desired frequency.
- We also have Voltmeter & Ammeter for both AC & DC.
- We will discuss them in detail in our coming lectures.

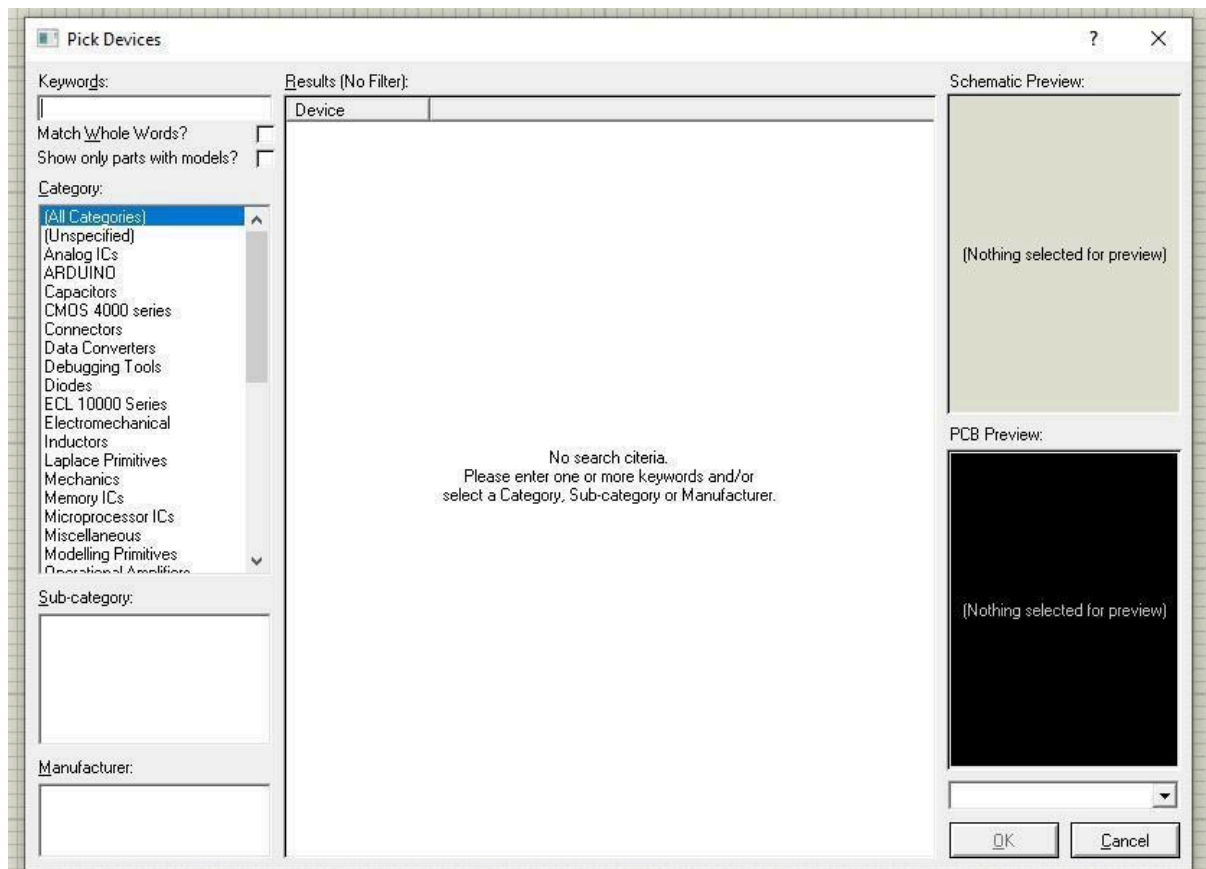


- As you can see in figure on right side, Icon A is called Graph mode, used to create graphs of voltage and current. It has different style of graphs.
- Icon B and C are voltage and current probes respectively. Suppose you have designed some circuit in Proteus and you want to check the value of voltage at any point in the circuit. In order to do so, simply select this voltage probe and place it there and when you run your circuit, the probe will show the value of voltage above it and same for current probe.
- Icon D is used when we want to design our own component in Proteus.
- Icon E is a simple text editor, used for placing labels, warning or components names etc.

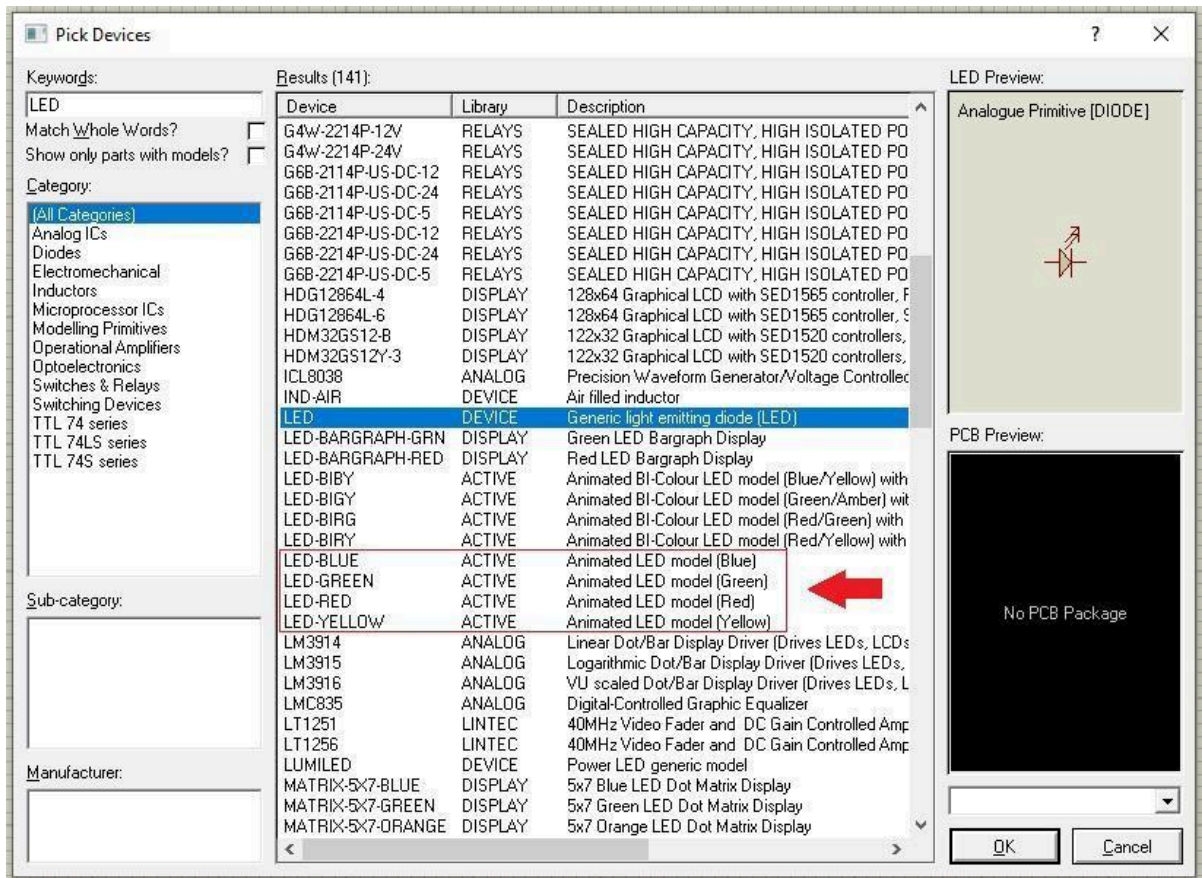
Simulate First Electronics Project in Proteus ISIS



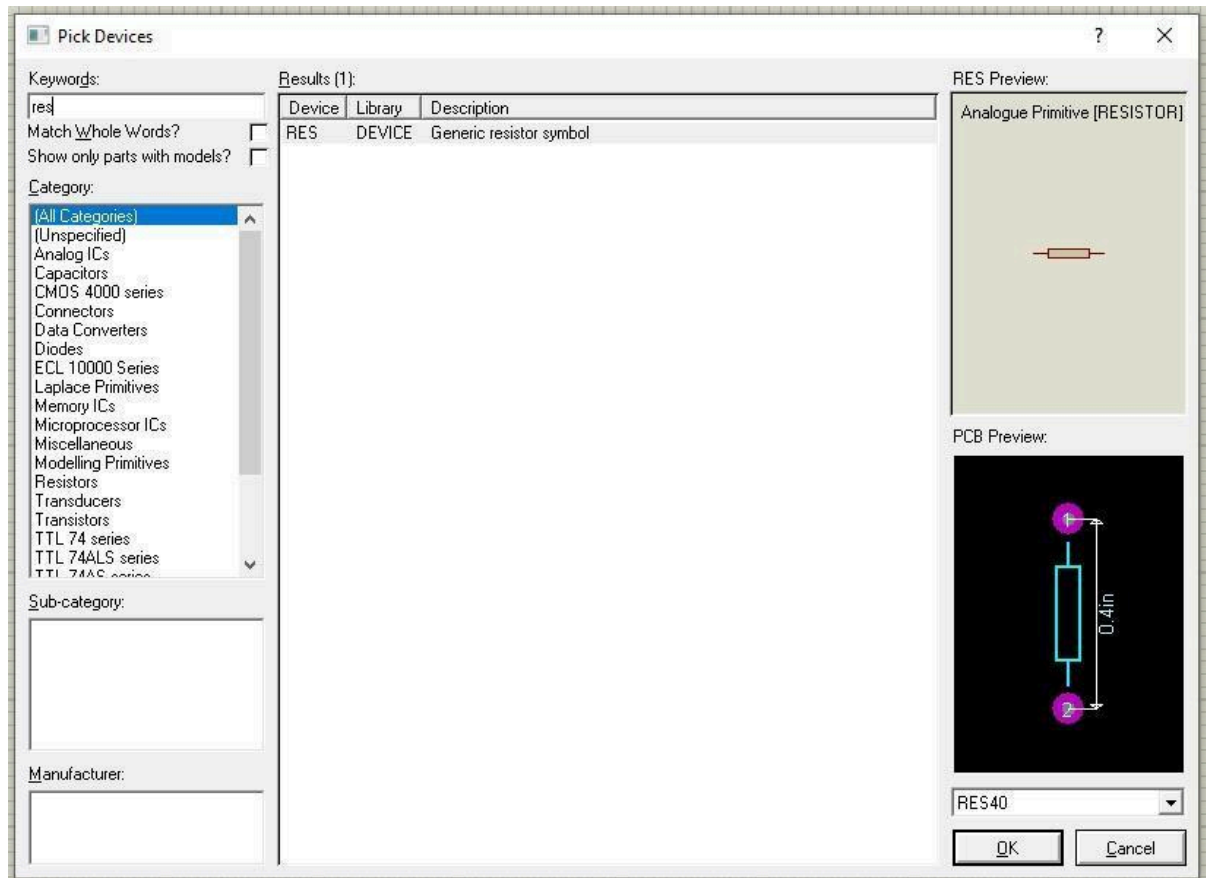
- Open your Proteus ISIS software and then click on Components Mode in left menu tab, as shown in the figure.
- After that click on the **P (Pick from Libraries)** Button, and a new window will open up.
- This new window is called **Pick Devices** and is used to make search for electronics components.
- Proteus has a huge database of electronics components in the form of libraries. i.e. Diode Library will have all the diode components.
- So, from these millions of components, we need to make a search for our required components to design electronic circuit.
- You can see Pick Devices window in below figure, so let's first discuss its layout:



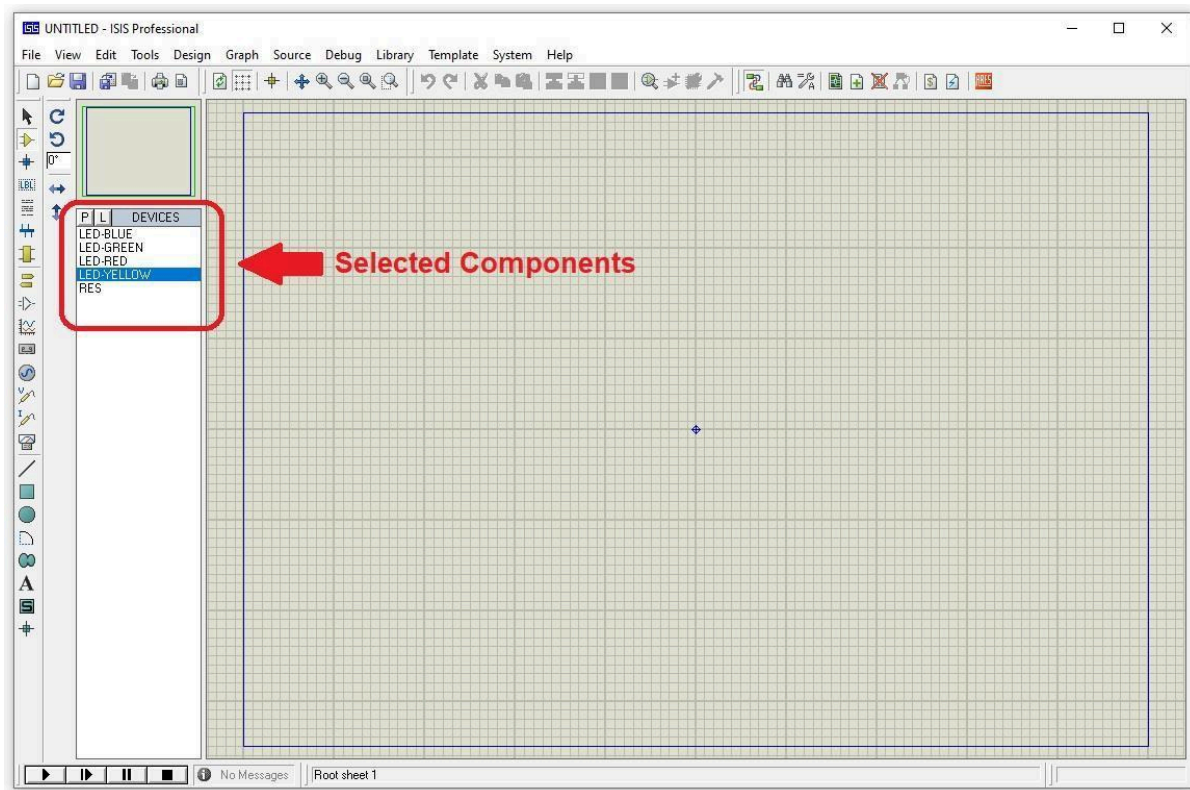
- **Keywords** textbox is used to make a search for any component and Proteus will display the related components in Results panel. (We will search in a while)
- **Category** Section displays all the categories available in Proteus and when you click on any category then it's components will be displayed in Results panel.
- After that, we have **Sub-category & Manufacturer**, rite now I don't have any.
- On the right side we have **Schematic Preview & PCB Review**, so when we select any component then its respective Previews will be shown here.
- So, now let's make a search for LED, as shown in below figure:



- As you can see in above figure that Proteus has provided us with 141 Results and I have boxed four LEDs, which I am going to use in my circuit.
- Moreover, Category section is now showing only those categories which are related to searched keyword.
- Moreover, we also have a Schematic Preview but we don't have any PCB Preview as it's not available for this component.
- So, double click on these four LEDs and they will be added in Proteus workspace.
- Moreover, we also need to add resistance so make a search for resistance, as shown in below figure:



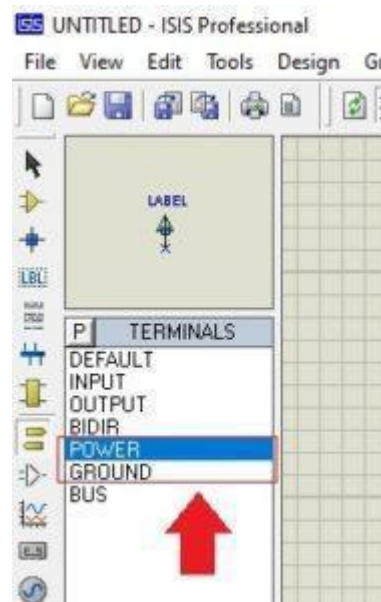
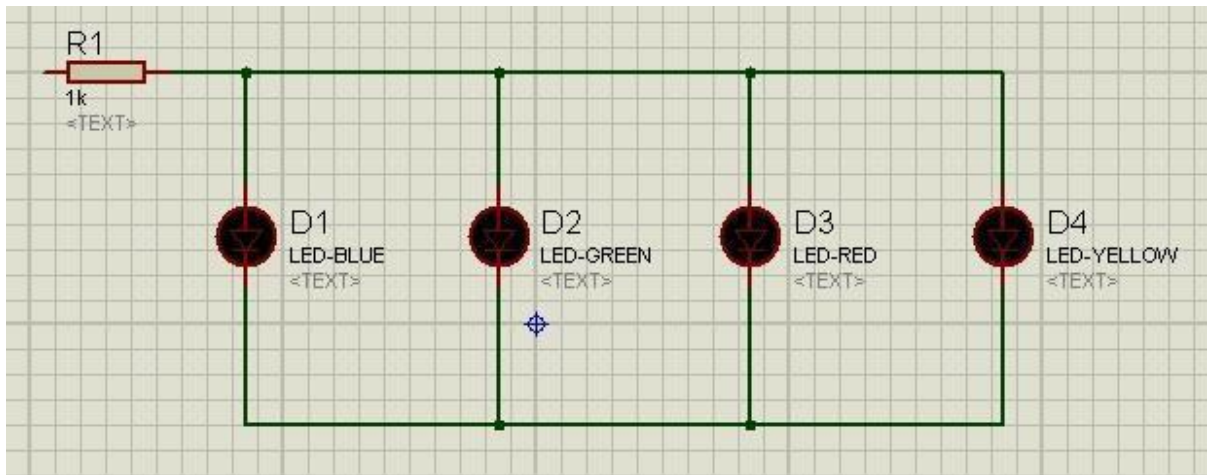
- Double click on this RES component and then close this **Pick Devices** window.
- You will get these selected components in the Devices section, as shown in below figure:



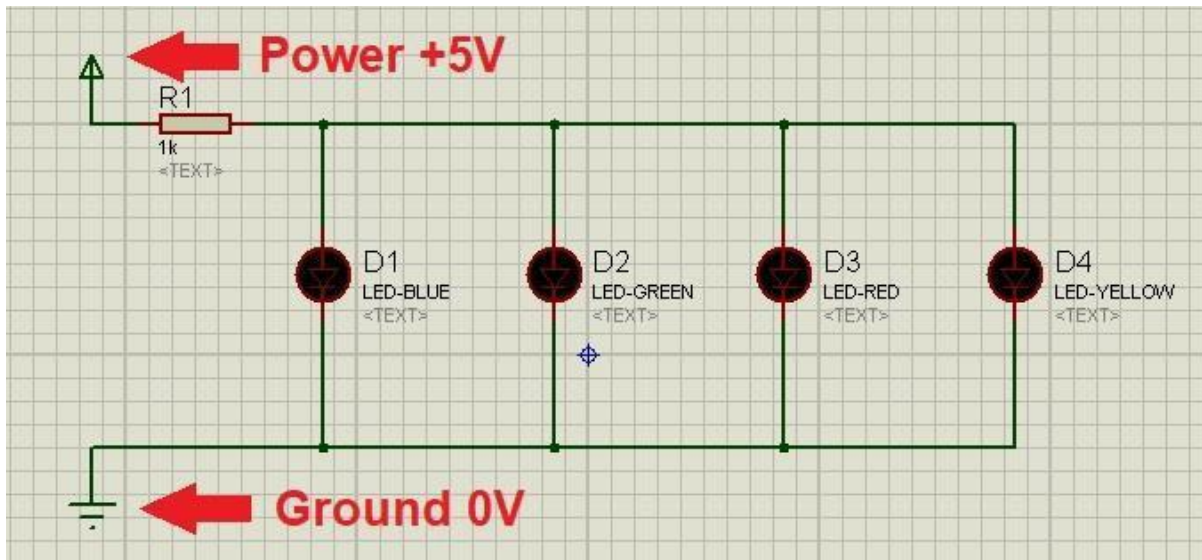
- As we are designing a simple project so we have selected just four components but in complex projects, we have a long list of components in this Devices section and it proves quite helpful.
- So, let's place these components, one by one in the central work area.



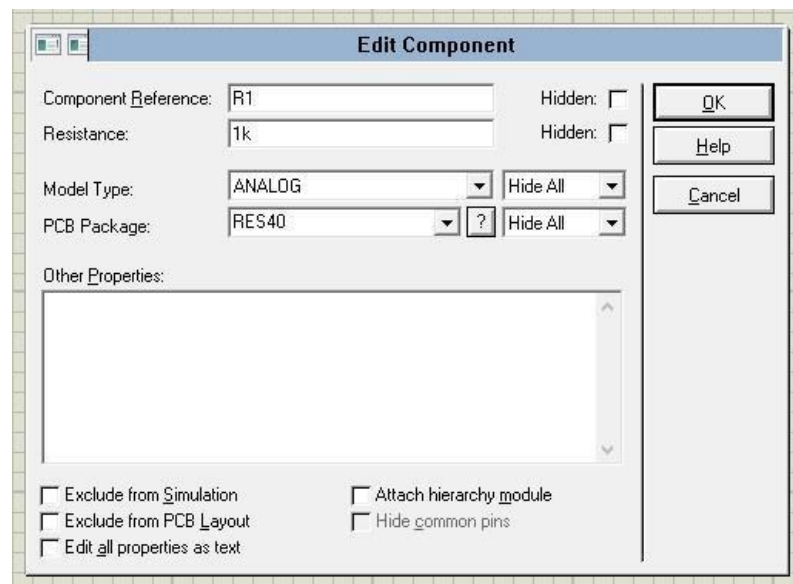
- You can drag & drop them OR can select by clicking and then again click to place.
- I have placed these components in the work area, as shown in figure on right side.
- So, now let's connect them together using wires and for that, we need to click on the pin terminal of each component.
- I have combined these electronic components together using wires, as shown in below figure:



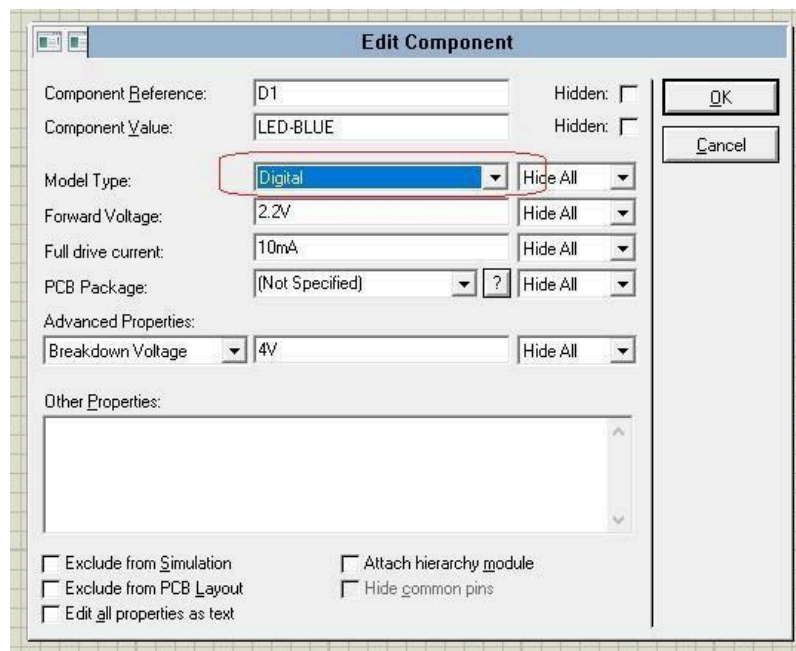
- Now we need to provide voltage supply to this circuit and there are several voltage sources in Proteus. (We will cover them in coming lectures)
- For now, let's click on the Terminals Mode in the left Toolbar and you will get Proteus Terminals, as shown in figure on right side.
- From these terminals, we are going to use **Power & Ground**, so place them in the circuit, as shown in below figure:
- We will discuss all these Terminal Components in detail in our coming lectures.



- If we place multiple Ground components in the circuit then Proteus will consider them all as connected/short.
- These Terminals are quite helpful, as in complex circuits, these wires can become too messy and we can avoid them by using these terminals.
- So, we have completely designed our circuit but we need to change the properties of these components a little.
- So, double click on resistance to open its Properties Panel, as shown in below figure:

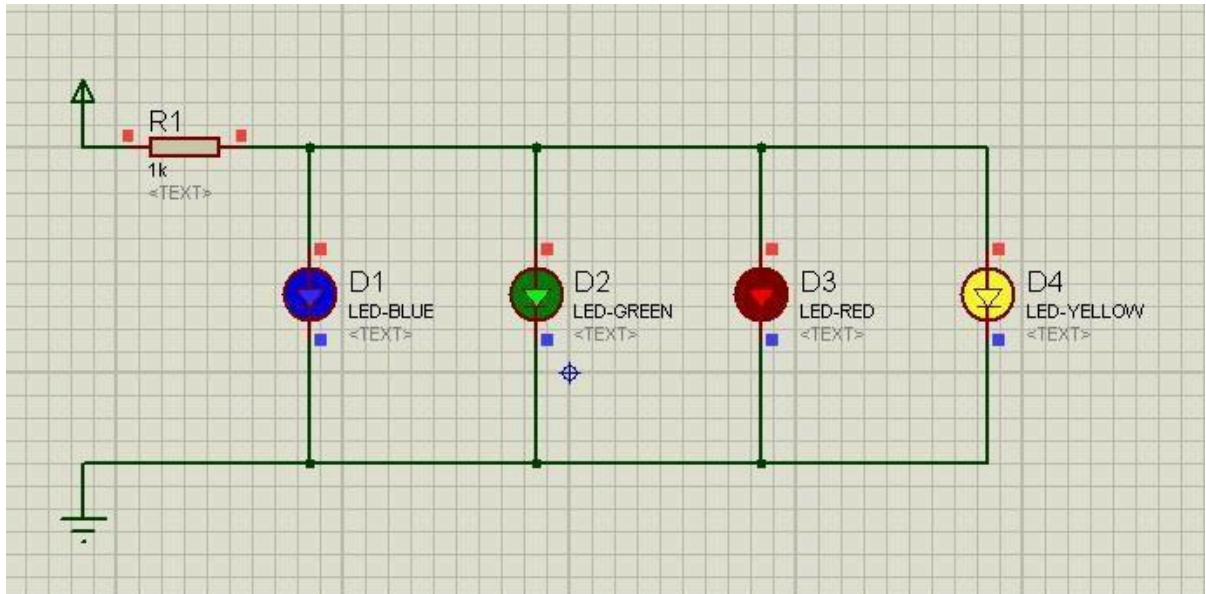


- From this Edit Component window, we can edit different properties of selected component.
- As you can see, first we have **Component Reference**, that's the name of our component i.e. R1. If we have multiple resistances, then there names will be R2, R3 and so on.
- We can't have multiple items with same Component Reference, as it will create an error.
- Second Property defines the resistance of the component and I have changed it from 10k to 1k.
- Then we have Model Type and its analog.
- Finally we have PCB Package, we will use it when we will be designing the PCB design of this circuit.
- So, click on the OK Button and resistance value will change from 10k to 1k.
- Now, double click on first LED to open its Properties Panel, as shown in below figure:



- As LED is a bit complex component as compared to resistance, that's why it has a lot more Properties to Edit.
- As we are designing a digital circuit, so we need to change the Model Type of LED from Analog to Digital and then click on the OK Button.
- You need to change this Model Type for all these four LEDs.
- So, now we have completely designed our first electronic circuit in Proteus.

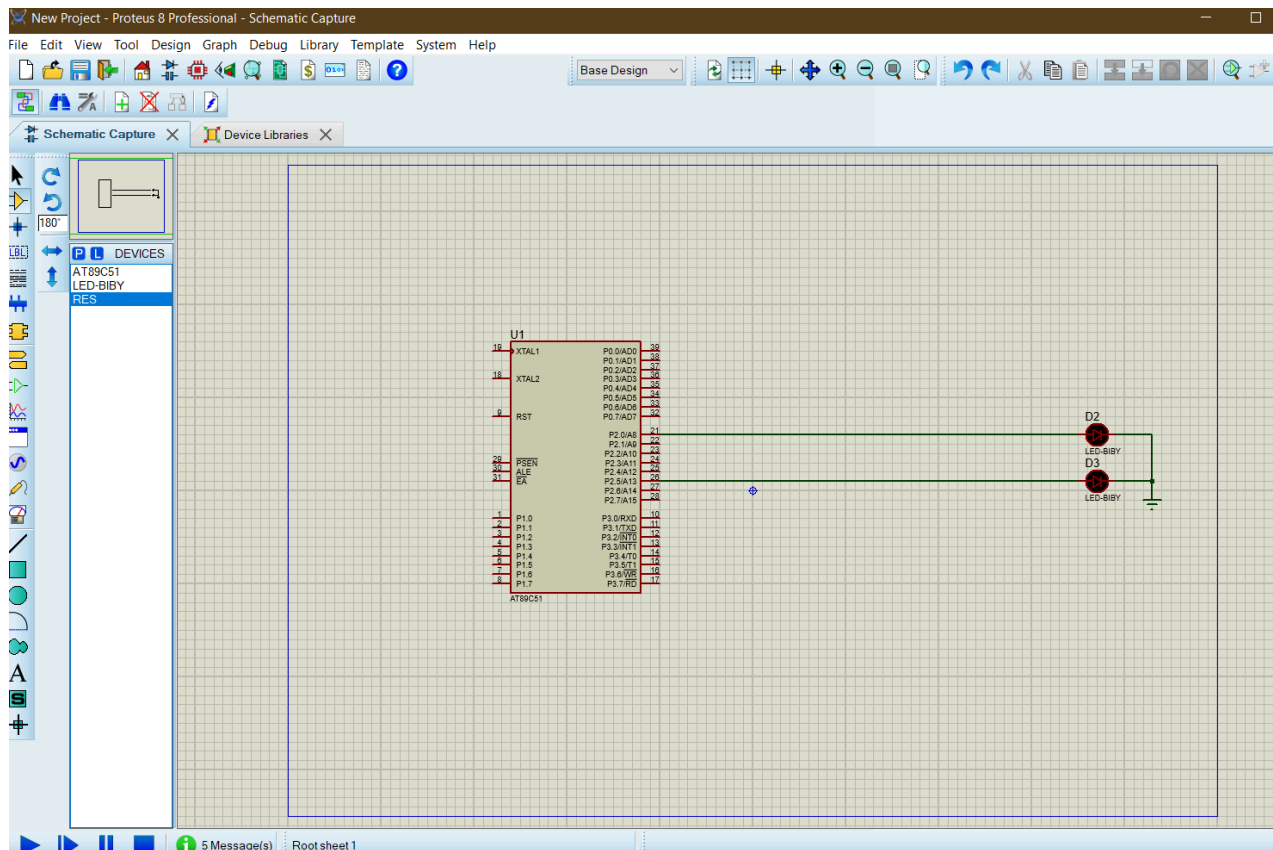
- Let's run this simulation, by clicking the Play button at the bottom.
- If everything goes fine, then all LEDs will glow, as shown in below figure:



- We have successfully simulated our first electronics circuit in Proteus ISIS and you can see these LEDs have different colors as specified in their Reference Value.

LED Blinking Project Using 8051 Microcontroller in Proteus ISIS

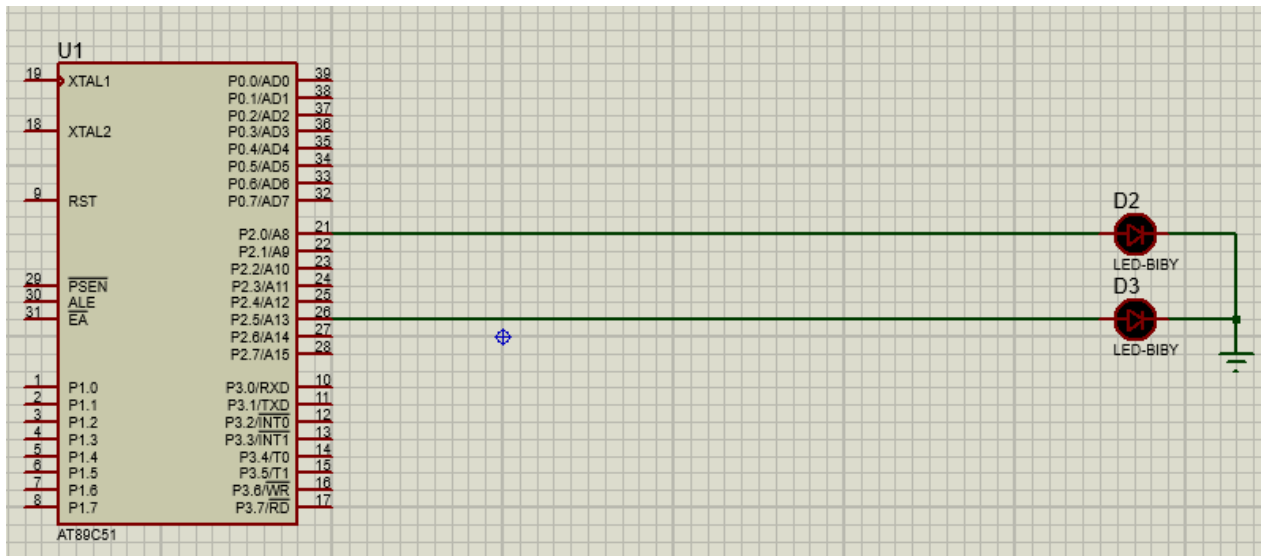
- let's first design the Proteus Simulation for LED Blinking Project Using 8051 Microcontroller.
- So, get these components from Proteus components library and place it in your workspace, these components are shown in below figure:



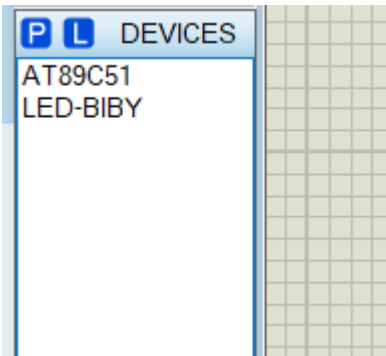
- Upload the 1.hex file given to you into your 8051 Microcontroller which I have used is AT89C51 and hit the RUN button.

PCB Designing in Proteus ARES

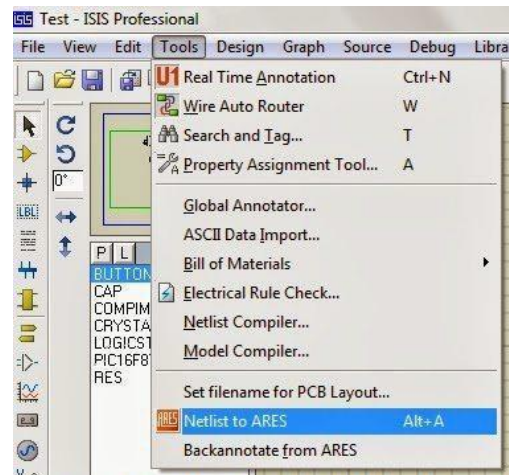
- As in this tutorial, I just want to give you an idea of How to design PCB that's why I haven't taken difficult circuit, just a simple PIC basic circuit.
- First design your circuit in Proteus ISIS as shown in below figure:



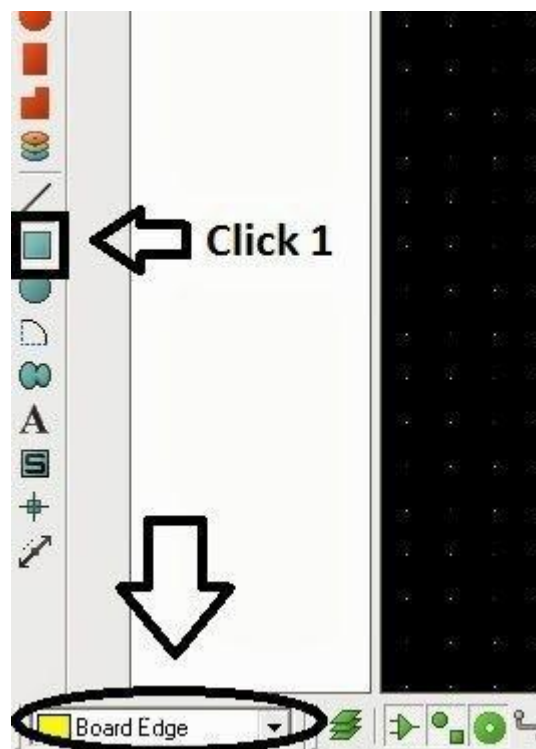
- These components are used while designing this simulation:



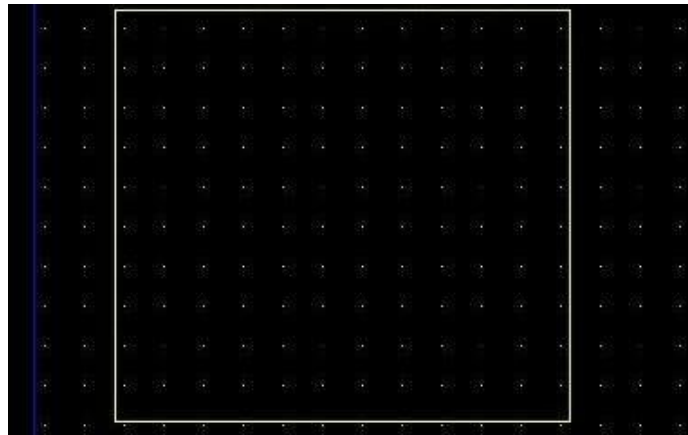
- After you got sure that your circuit is perfect and ready for designing, then click on the **Tools** and then **Netlist to ARES** as shown below:



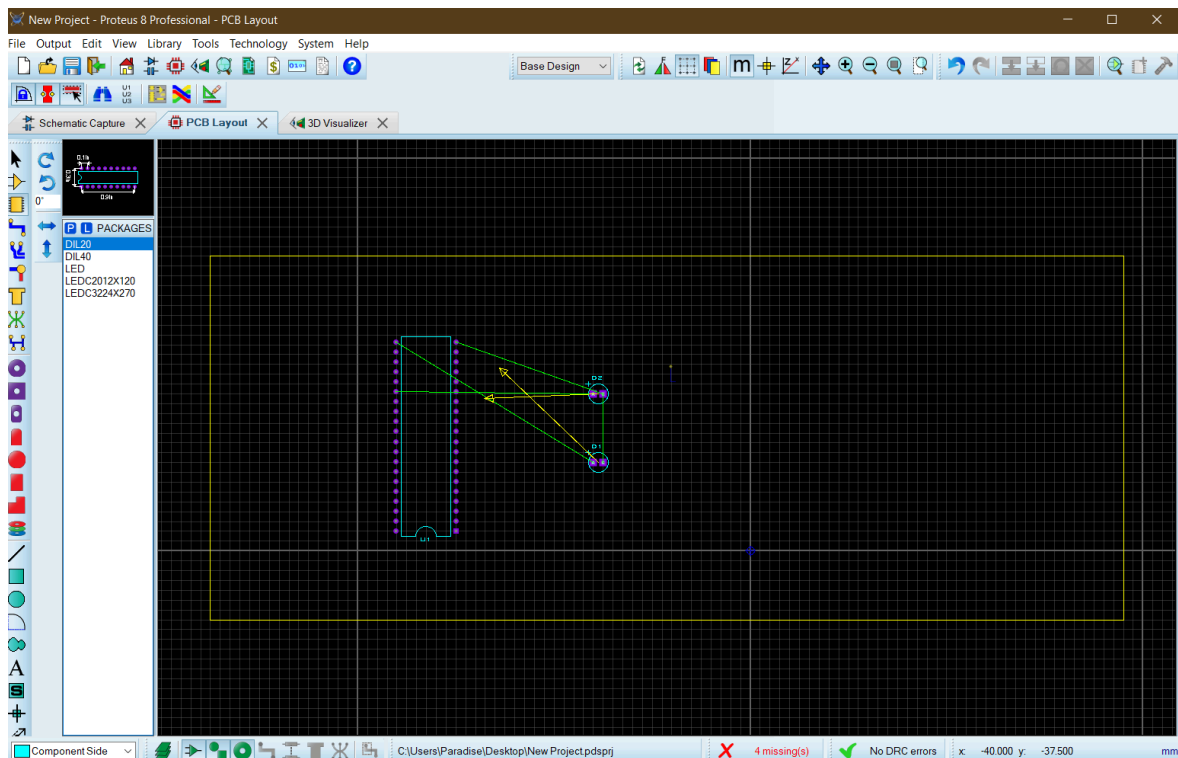
- After clicking, Proteus ARES will be opened.
- Now in Proteus ARES, select the block option from left toolbar and also make sure that you selected **Board Edge** in the below drop down menu as shown in below figure:



- Now make a rectangular block in the workspace, this block is actually the boundary of your PCB.
- You can set its proper dimensions and can also re-size it manually using the mouse.



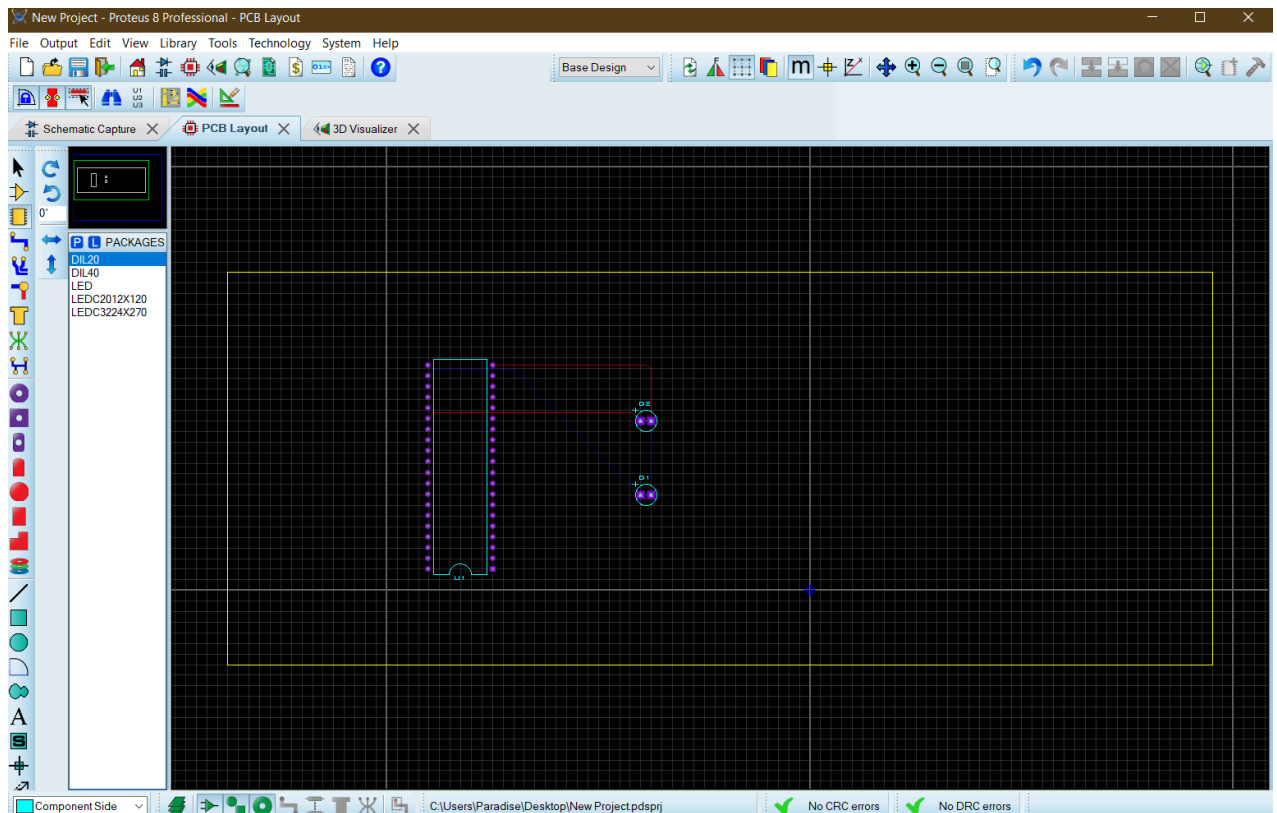
- Now select the component option from the left toolbar, it will show all the components used in your circuit.
- Place all these components in the workspace one by one as shown below. These green lines shown in the below image is actually the software intelligence.
- Using the circuit, it gives us the routes automatically and we don't need to panic any more just need to follow these route, if we are doing manually routing.



- Now there are two ways of adding routing, first method is auto routing.
- To do auto routing, click on **Tools** and then **Auto Router** and a property box will open where you can set many different option for routing like the width of route and the PCB layers etc.
- After selecting your properties just click on **Begin Routing**.



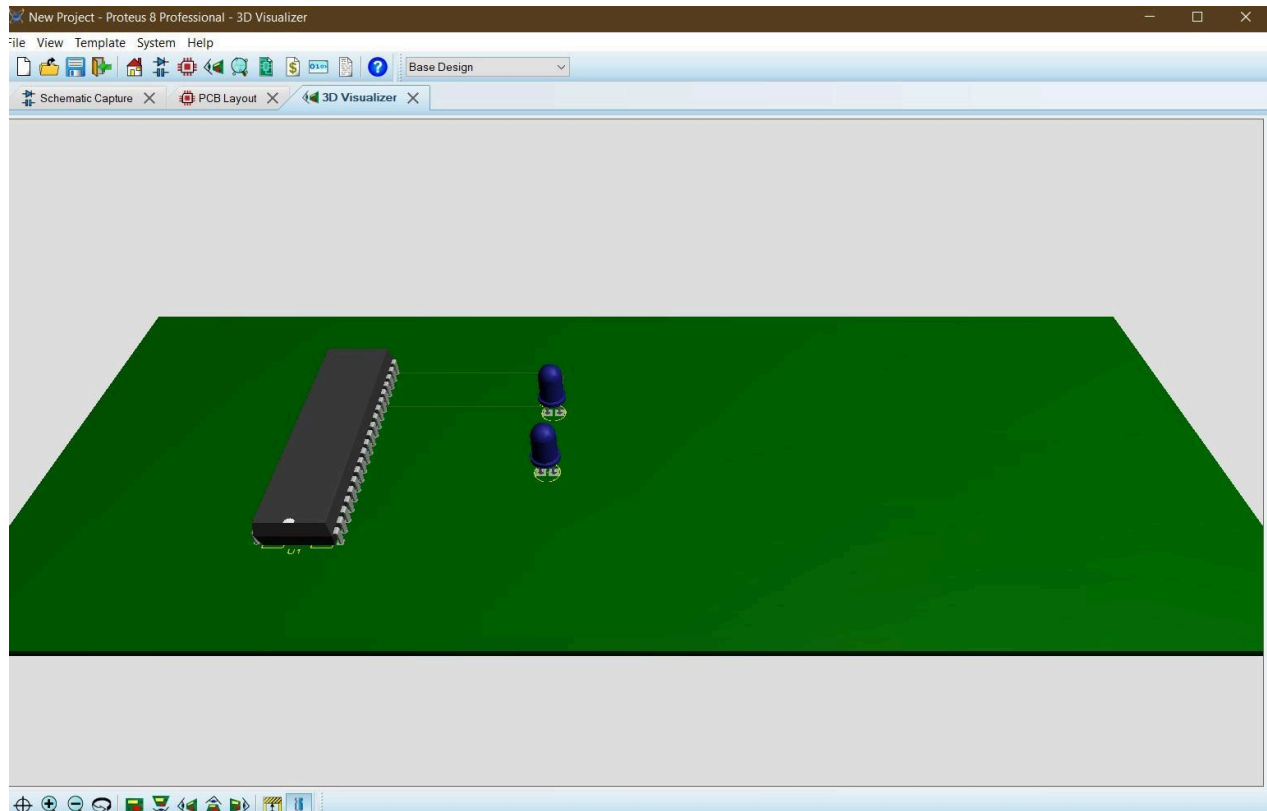
- And then a magic will start and you PCB will become ready as shown in below figure:



- Second method is manual routing, you can do manual routing by clicking the edges just like we connect wires in Proteus ISIS. For manual routing select the option shown in below figure and start routing.

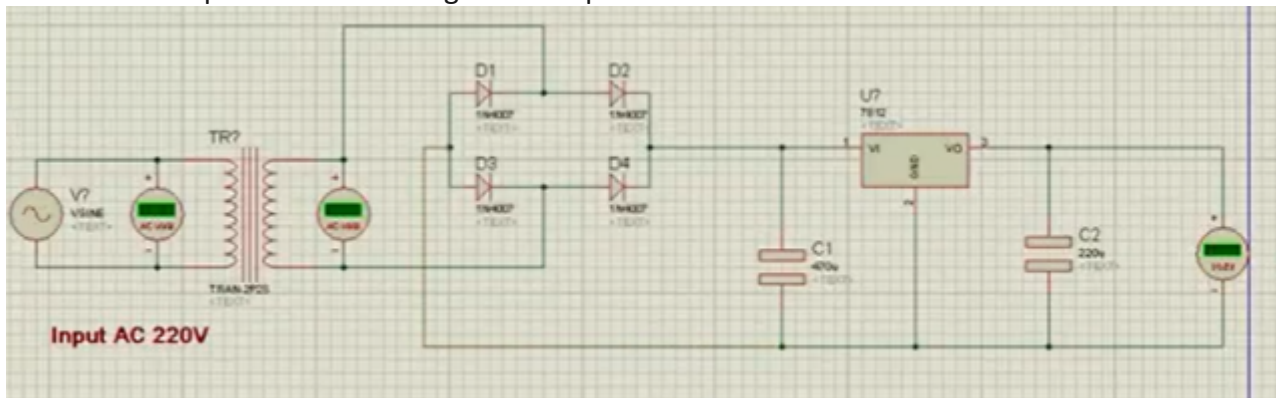
That's it. I don't think its much difficult. I am gonna stop this tutorial here. I have tried my best to share my knowledge about Proteus. IF you guys having any problem in part of this tutorial, feel free to contact me. Take care.

After you complete tracking, save your project in the same file you saved your Proteus project. You can choose 3D visualization o see the final circuit. You can view all the angles, components, board without the components placed, front and back view.

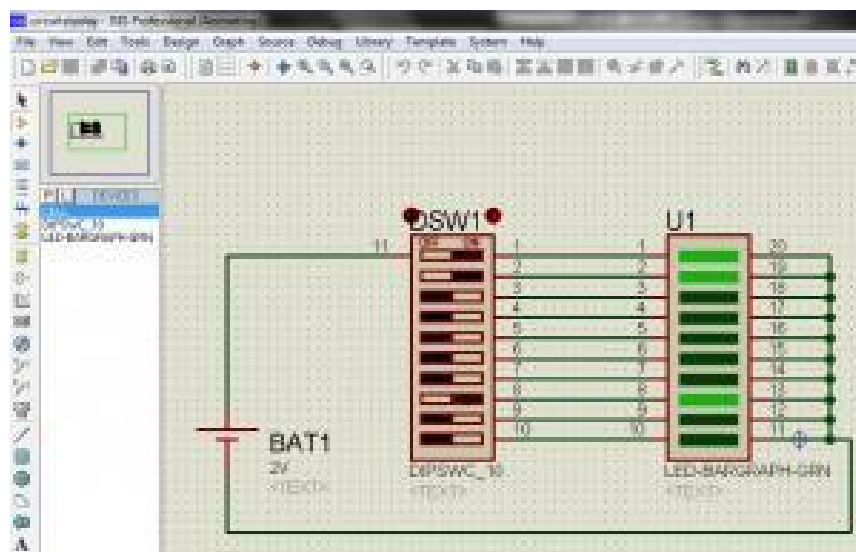


Lab Task 1:

Show the output of the following circuit in proteus ISIS:

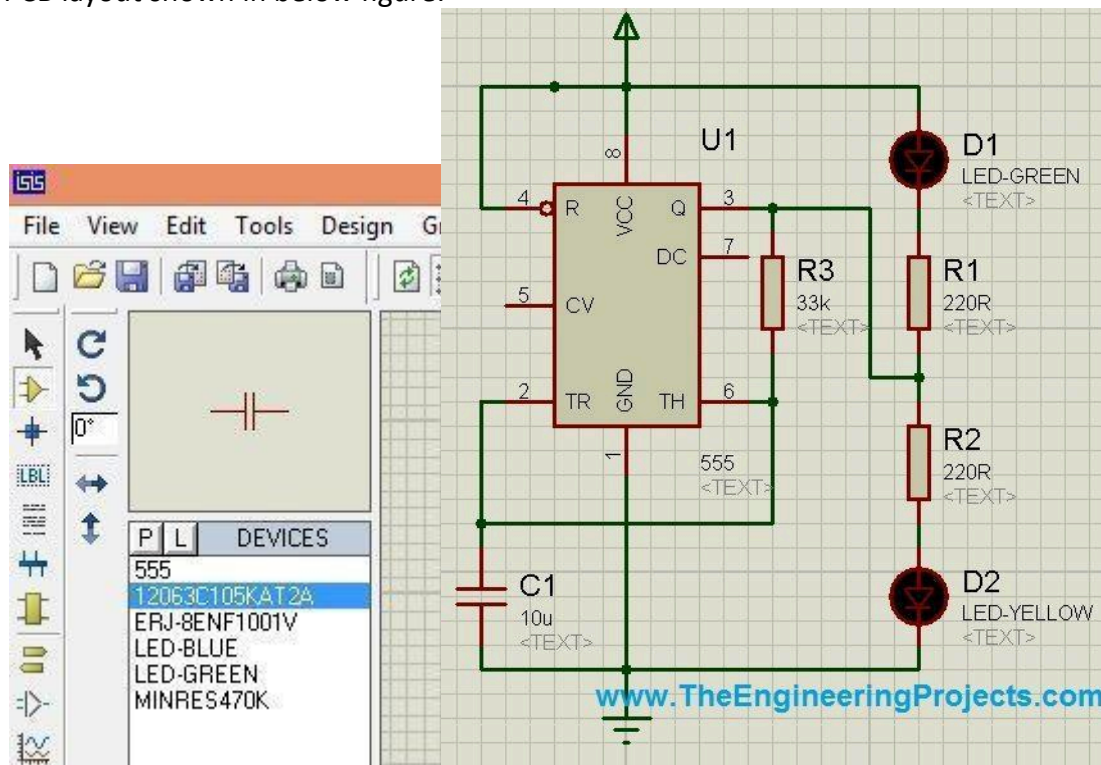


Department of Computer Engineering
EC-442 Embedded System



Lab Task 2:

Show the output by flashing two LEDs with 555 Timer in Proteus ISIS by designing a circuit as well PCB layout shown in below figure:



Lab Performance Evaluation:

Read and practice the manual. Performance is evaluated at the end of lab based on successfully running and understanding of examples, tasks and viva using defined rubrics.

Lab Report Evaluation:

Submit the Lab Report by writing all the examples and tasks which must include the following:

1. Brief description (3-5 lines)
 2. The steps
 3. The results (Screenshot)
- Task 2 is evaluated in lab report.

Important note: Lab Report must be according to the Lab Report Format available on Google Classroom and must be submitted on time. Two similar reports will get zero marks. So, don't try to copy your fellow reports. Lab report must be submitted in group of three maximum.